



Fonds National de la
Recherche Luxembourg



MINIMAL MODEL FOR THERMODYNAMICS OF NONEQUILIBRIUM PHASE TRANSITIONS

Massimiliano Esposito

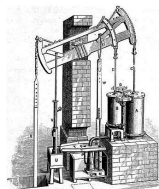
Tim Herpich and Juzar Thingna

Université du Luxembourg

Luxembourg, May 15, 2018

Critics EU - CriTiCS DTU Lux meeting

19th century thermodynamics:

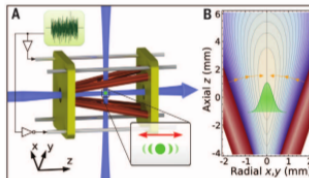
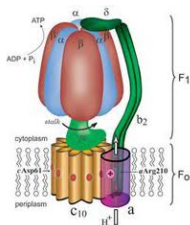


No dynamics

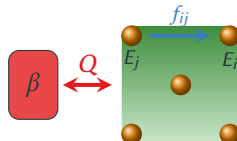
21st century thermodynamics:

Small $\rightarrow$ $\left\{ \begin{array}{l} \text{Fluctuations} \\ \text{Far-from-equilibrium} \\ \text{Quantum effects} \end{array} \right.$

Build thermodynamics on dynamics



- **System** with **states** i of energy E_i and occupation probability p_i interacting with a **heat bath** at inverse temperature β and subjected to a **nonconservative force** f .



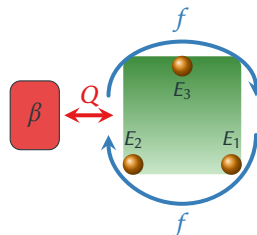
- **Markovian master equation** and **local detailed balance**

$$\dot{p}_i = \sum_j W_{ij} p_j, \quad \ln \frac{W_{ij}}{W_{ji}} = -\beta (E_i - E_j - f_{ij})$$

- **First law of thermodynamics:** $d_t \langle E \rangle = \langle \dot{W} \rangle + \langle \dot{Q} \rangle$
 - Energy current $d_t \langle E \rangle = d_t \sum_i E_i p_i = \sum_{i,j} (E_i - E_j) W_{ij} p_j$
 - Work current $\langle \dot{W} \rangle = \sum_{i,j} f_{ij} W_{ij} p_j$
 - Heat current $\langle \dot{Q} \rangle = \sum_{i,j} (E_i - E_j - f_{ij}) W_{ij} p_j = -\frac{1}{\beta} \sum_{i,j} W_{ij} p_j \ln \frac{W_{ij}}{W_{ji}}$
- **Second law of thermodynamics:** $\langle \dot{S} \rangle = d_t \langle S \rangle - \langle \dot{S}_e \rangle \geq 0$
 - Nonequilibrium entropy $\langle S \rangle = -\sum_i p_i \ln p_i$
 - Entropy flow $\langle \dot{S}_e \rangle = -\sum_{i,j} W_{ij} p_j \ln \frac{W_{ij}}{W_{ji}} = \beta \langle \dot{Q} \rangle$
 - Entropy production $\langle \dot{S} \rangle = \sum_{i,j} W_{ij} p_j \ln \frac{W_{ij} p_j}{W_{ji} p_i} \geq 0$

EXAMPLE: DRIVEN THREE-STATE UNIT

Three-state unit [PRL 96, 145701 (2006)] with **states** 1, 2, 3 of energy E_1, E_2, E_3 and occupation probability p_1, p_2, p_3 interacting with a **heat bath** at inverse temperature β subjected to a **nonconservative rotational forcing** f .



- **Markovian master equation** and **local detailed balance**

$$\dot{p}_i = \sum_{j=1}^3 W_{ij} p_j, \quad \ln \frac{W_{ij}}{W_{ji}} = -\beta (E_i - E_j - f \Theta(i, j)), \quad \Theta(i, j) = \begin{cases} 1 & (i - j) \bmod 3 = 1 \\ -1 & \text{else} \end{cases}$$

- **First law** of thermodynamics: $d_t \langle E \rangle = \langle \dot{W} \rangle + \langle \dot{Q} \rangle$

- Energy current $d_t \langle E \rangle = d_t \sum_{i=1}^3 E_i p_i = \sum_{i,j=1}^3 (E_i - E_j) W_{ij} p_j$
- Work current $\langle \dot{W} \rangle = f \sum_{i,j=1}^3 \Theta(i, j) W_{ij} p_j$
- Heat current $\langle \dot{Q} \rangle = \sum_{i,j=1}^3 (E_i - E_j - f \Theta(i, j)) W_{ij} p_j = \sum_{i,j=1}^3 -\frac{1}{\beta} \ln \left(\frac{W_{ij}}{W_{ji}} \right) W_{ij} p_j$

- **Second law** of thermodynamics: $\langle \dot{\Sigma} \rangle = d_t \langle S \rangle - \langle \dot{S}_e \rangle \geq 0$

- **Detailed fluctuation theorems:**

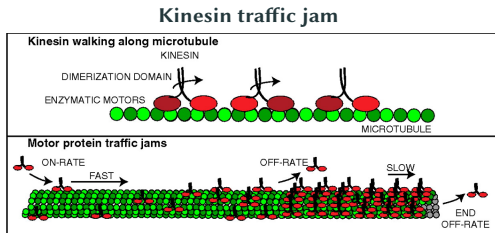
Crooks $\frac{P(W = w)}{P(W = -w)} = e^{\beta w}$

Seifert $\frac{P(\Sigma = \sigma)}{P(\Sigma = -\sigma)} = e^{\sigma}$

INTRODUCTION AND MOTIVATION

- **Nonequilibrium phase transitions** are most commonly discussed in the framework of nonlinear dynamics and bifurcation theory.
- **Stochastic thermodynamics** provides a systematic treatment of nonequilibrium thermodynamics $\Rightarrow$ **thermodynamics of nonequilibrium phase transitions**.
- **Practical motivation** to explore the thermodynamics of nonequilibrium phase transitions in large ensembles of interacting degrees of freedom:
 - Limitations of energy-converting nanomachines $\Rightarrow$ Consider ensembles of nanomachines
 - Phase transitions affect performance of large ensembles of machines?
 - Role of the number of nanomachines?

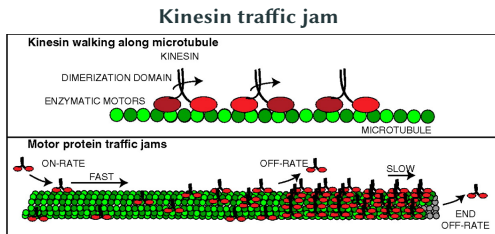
- **Example:**



- **Synchronization** as a promising candidate for a beneficial performance of the ensemble.

INTRODUCTION AND MOTIVATION

- **Nonequilibrium phase transitions** are most commonly discussed in the framework of nonlinear dynamics and bifurcation theory.
- **Stochastic thermodynamics** provides a systematic treatment of nonequilibrium thermodynamics $\Rightarrow$ **thermodynamics of nonequilibrium phase transitions**.
- **Practical motivation** to explore the thermodynamics of nonequilibrium phase transitions in large ensembles of interacting degrees of freedom:
 - Limitations of energy-converting nanomachines $\Rightarrow$ Consider ensembles of nanomachines
 - Phase transitions affect performance of large ensembles of machines?
 - Role of the number of nanomachines?
- **Example:**

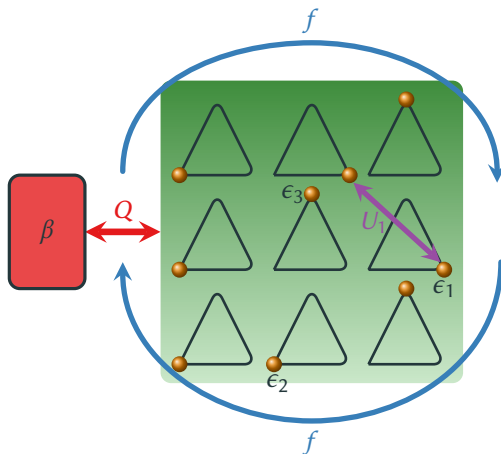


- **Synchronization** as a promising candidate for a beneficial performance of the ensemble.

$\Rightarrow$ Search for a (feasible) thermodynamic model exhibiting synchronization.

OPEN THREE-STATE UNIT NETWORK

- **Network** of N coupled three-state units.
- Open system in contact with **heat bath**.
- Microstate: $\alpha = (2, 1, 3, 2, 3, 1, 2, 2, 3)$
- **Interaction energy** $U(\alpha) = \frac{u}{2N} \sum_{k=1}^3 N_k^2(\alpha) + U_0$.
- Subjected to **torque** f acting as a global bias.
- Dimension of state space $\mathcal{D}(\alpha) = 3^N - 1$.



- Markovian master equation with Arrhenius rates

$$\dot{p}_\alpha = \sum_{\alpha'} w_{\alpha\alpha'} p_{\alpha'}, \quad w_{\alpha\alpha'} = \Gamma e^{-\frac{\beta}{2}(\Delta E(\alpha, \alpha') - \Theta(\alpha, \alpha')f)}, \quad \Theta(\alpha, \alpha') = \begin{cases} 1 & \text{if } \sum_{i=1}^N (\alpha_i - \alpha'_i) = 1 \bmod 3 \\ -1 & \text{else} \end{cases}$$

- Local detailed balance

$$\ln \frac{w_{\alpha\alpha'}}{w_{\alpha'\alpha}} = -\beta (\Delta E(\alpha, \alpha') - \Theta(\alpha, \alpha')f), \quad \Delta E(\alpha, \alpha') = \sum_{i=1}^3 \left\{ [N_i(\alpha) - N_i(\alpha')] \epsilon_i + \frac{u}{2N} [N_i(\alpha)^2 - N_i(\alpha')^2] \right\}.$$

Thermodynamics

First law

$$\begin{aligned} d_t \langle e \rangle &\equiv \sum_{\alpha, \alpha'} E(\alpha) w_{\alpha\alpha'} p_{\alpha'} = \langle \dot{q} \rangle + \langle \dot{w} \rangle \\ \langle \dot{q} \rangle &\equiv \sum_{\alpha, \alpha'} [E(\alpha) - f \Theta(\alpha, \alpha')] w_{\alpha\alpha'} p_{\alpha'} \\ \langle \dot{w} \rangle &\equiv \sum_{\alpha, \alpha'} f \Theta(\alpha, \alpha') w_{\alpha\alpha'} p_{\alpha'}, \end{aligned}$$

Second law

$$\begin{aligned} d_t \langle s \rangle &\equiv d_t \sum_{\alpha} p_{\alpha} \ln p_{\alpha} = \langle \dot{s}_e \rangle + \langle \dot{\sigma} \rangle \\ \langle \dot{s}_e \rangle &\equiv - \sum_{\alpha, \alpha'} w_{\alpha\alpha'} p_{\alpha'} \ln \frac{w_{\alpha\alpha'}}{w_{\alpha'\alpha}} = \beta \langle \dot{q} \rangle \\ \langle \dot{\sigma} \rangle &\equiv \sum_{\alpha, \alpha'} w_{\alpha\alpha'} p_{\alpha'} \ln \frac{w_{\alpha\alpha'} p_{\alpha'}}{w_{\alpha'\alpha} p_{\alpha}} \geq 0. \end{aligned}$$

- Using $P_N \equiv \sum_{\alpha \in N} p_\alpha$ with mesostate $\mathbf{N} = (N_1, N_2)$, leads to **mesoscopic master equation**

$$\dot{P}_N = \sum_{N'} W_{N,N'} P_{N'}, \quad \mathcal{D}(\mathbf{N}) = \frac{(N+1)(N+2)-2}{2} \Big] \stackrel{N \rightarrow \infty}{\sim} \frac{N^2}{2}$$

with mesoscopic transition rates satisfying **local detailed balance**

$$\ln \frac{W_{\mathbf{N}\mathbf{N}'}}{W_{\mathbf{N}'\mathbf{N}}} = -\beta [\Delta F^{eq}(\mathbf{N}, \mathbf{N}') - \Theta(\mathbf{N}, \mathbf{N}')(\mathbf{N}, \mathbf{N}')f], \quad F^{eq}(\mathbf{N}) = E(\mathbf{N}) - \beta^{-1} \underbrace{\ln \frac{N!}{\prod_i N_i!}}_{S^{int}(\mathbf{N})}$$

Thermodynamics

First law

$$d_t \langle E \rangle \equiv \sum_{\mathbf{N}, \mathbf{N}'} E(\mathbf{N}) W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'} = \langle \dot{Q} \rangle + \langle \dot{W} \rangle$$

$$\langle \dot{Q} \rangle \equiv \sum_{\mathbf{N}, \mathbf{N}'} [E(\mathbf{N}) - f \Theta(\mathbf{N}, \mathbf{N}')] W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'}$$

$$\langle \dot{W} \rangle \equiv \sum_{\mathbf{N}, \mathbf{N}'} f \Theta(\mathbf{N}, \mathbf{N}') W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'},$$

Second law

$$d_t \langle S \rangle \equiv d_t \sum_{\mathbf{N}} P_{\mathbf{N}} [S^{int}(\mathbf{N}) - \ln P_{\mathbf{N}}] = \langle \dot{S}_e \rangle + \langle \dot{\Sigma} \rangle$$

$$\langle \dot{S}_e \rangle \equiv - \sum_{\mathbf{N}, \mathbf{N}'} W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'} \left[\ln \frac{W_{\mathbf{N}\mathbf{N}'}}{W_{\mathbf{N}'\mathbf{N}}} + \Delta S^{int}(\mathbf{N}, \mathbf{N}') \right] = \beta \langle \dot{Q} \rangle$$

$$\langle \dot{\Sigma} \rangle \equiv \sum_{\mathbf{N}, \mathbf{N}'} W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'} \ln \frac{W_{\mathbf{N}\mathbf{N}'}}{W_{\mathbf{N}'\mathbf{N}} P_{\mathbf{N}}} \geq 0.$$

- Equivalence** of the **first law** observables: $d_t \langle e \rangle = d_t \langle E \rangle$, $\langle \dot{q} \rangle = \langle \dot{Q} \rangle$, $\langle \dot{w} \rangle = \langle \dot{W} \rangle$, $\langle \dot{s}_e \rangle = \langle \dot{S}_e \rangle$.
- Inequivalence** of system entropy and entropy production: $d_t \langle s \rangle \neq d_t \langle S \rangle$, $\langle \dot{\sigma} \rangle \neq \langle \dot{\Sigma} \rangle$.

But: $d_t \langle s^S \rangle = d_t \langle S^S \rangle = 0 \Rightarrow \langle \dot{\sigma}^S \rangle = \langle \dot{\Sigma}^S \rangle = -\langle \dot{s}_e^S \rangle = -\langle \dot{S}_e^S \rangle \geq 0$.

- Mean-field approximation yields a **two-dimensional system of nonlinear differential equations**

$$\dot{\bar{n}}_i(t) \equiv \langle \dot{\bar{n}}_i(t) \rangle = \sum_j k_{ij}(\bar{n}_i, \bar{n}_j) \bar{n}_j(t)$$

with macroscopic transition rates satisfying **local detailed balance**

$$\ln \frac{k_{ij}}{k_{ji}} = -\beta [\epsilon_i - \epsilon_j + u(\bar{n}_i - \bar{n}_j) - \Theta(i, j)f] .$$

Thermodynamics

First law

$$d_t \langle \mathcal{E} \rangle \equiv \sum_{i,j} E_i k_{ij} \bar{n}_j = \langle \dot{Q} \rangle + \langle \dot{W} \rangle$$

$$\langle \dot{Q} \rangle \equiv \sum_{i,j} [E_i - f \Theta(i, j)] k_{ij} \bar{n}_j$$

$$\langle \dot{W} \rangle \equiv \sum_{i,j} f \Theta(i, j) k_{ij} \bar{n}_j ,$$

Second law

$$d_t \langle S \rangle \equiv -d_t \sum_i \bar{n}_i \ln \bar{n}_i = \langle \dot{S}_e \rangle + \langle \dot{S}_i \rangle$$

$$\langle \dot{S}_e \rangle \equiv - \sum_{i,j} k_{ij} \bar{n}_j \ln \frac{k_{ij}}{k_{ji}} = \beta \langle \dot{Q} \rangle$$

$$\langle \dot{S}_i \rangle \equiv \sum_{i,j} k_{ij} \bar{n}_i \ln \frac{k_{ij} \bar{n}_j}{k_{ji} \bar{n}_i} \geq 0 .$$

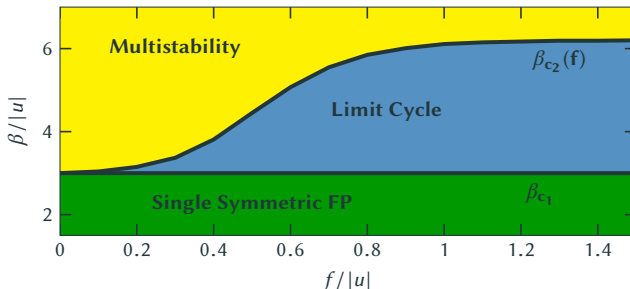
- Equivalence of the **first law observables**:

$$\lim_{N \rightarrow \infty} \frac{1}{N} d_t \langle E \rangle = d_t \langle \mathcal{E} \rangle, \quad \lim_{N \rightarrow \infty} \frac{1}{N} \langle \dot{Q} \rangle = \langle \dot{Q} \rangle, \quad \lim_{N \rightarrow \infty} \frac{1}{N} \langle \dot{W} \rangle = \langle \dot{W} \rangle .$$

- Equivalence of the **second law only** in the stationary state:

$$d_t \langle S^s \rangle = 0 \Rightarrow \lim_{N \rightarrow \infty} \frac{1}{N} \langle \dot{S}^s \rangle = - \lim_{N \rightarrow \infty} \frac{1}{N} \langle \dot{S}_e^s \rangle = \langle \dot{S}_i^s \rangle = - \langle \dot{S}_e^s \rangle \geq 0 .$$

- For $\epsilon_i = \epsilon$, the nonlinear equation has a **symmetric fixed point** $\dot{\bar{n}}_i = \sum_j k_{ij} \bar{n}_j |_{\bar{n}_j=1/3} = 0$.
- Linear stability indicates **Hopf bifurcation**: $\lambda_{\pm} = -\Gamma(\beta u + 3) \cosh\left(\frac{\beta f}{2}\right) \pm i\sqrt{3}\Gamma \sinh\left(\frac{\beta f}{2}\right)$.



- One equilibrium (first-order) and two nonequilibrium phase transitions at $\beta = \beta_{c1,2}$.
- Three distinct fixed points in the **low-temperature regime** ($\beta \gg \beta_c$) **Energy-driven**
- Symmetric solution in the **high-temperature regime** ($\beta \ll \beta_c$) **Entropy-driven**
- Thermodynamic consistency constraints **stable oscillations** to emerge only at intermediate temperatures ($\beta_{c1} < \beta \leq \beta_{c2}$).

$\Rightarrow$ Minimal thermodynamic model exhibiting synchronization

Stochastic Dynamics

- The microscopic/mesoscopic dynamics is ruled by an irreducible and Markovian master equation $\dot{P}_{\mathbf{N}} = \sum_{\mathbf{N}'} W_{\mathbf{N},\mathbf{N}'} P_{\mathbf{N}'}$ and exhibits a unique steady state (PF theorem).

Mean-Field Dynamics

- The macroscopic dynamics is governed by a nonlinear master equation $\dot{\bar{n}}_i = \sum_j k_{ij} \bar{n}_j$ and displays stable oscillations ($\beta_{c1} \leq \beta < \beta_{c2}$) and multistability ($\beta \geq \beta_{c2}$).

$\Rightarrow$ **Non-commutation of the thermodynamic- and the stationary limit.**

$$\lim_{t \rightarrow \infty} \lim_{N \rightarrow \infty} \mathbf{P}(t) \neq \lim_{N \rightarrow \infty} \underbrace{\lim_{t \rightarrow \infty} \mathbf{P}(t)}_{\mathbf{p}^s}, \quad \text{if } \beta \geq \beta_{c1}.$$

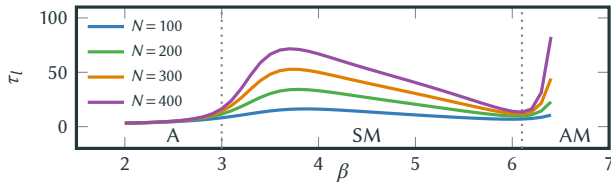
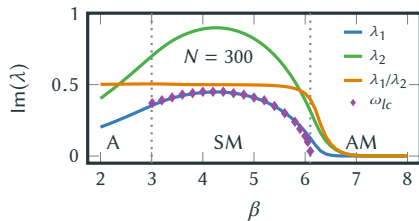
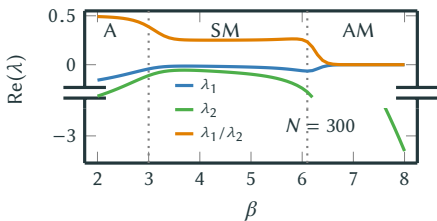
- Apparent paradox since mean field represents the exact solution in the $N \rightarrow \infty$ limit.
- Idea:** Infinite-time solutions of mean-field exhibited also by finite systems at long but *finite* timescales $\rightarrow$ **Metastability?**

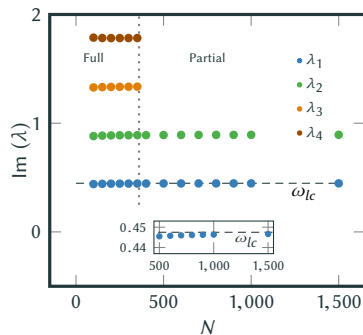
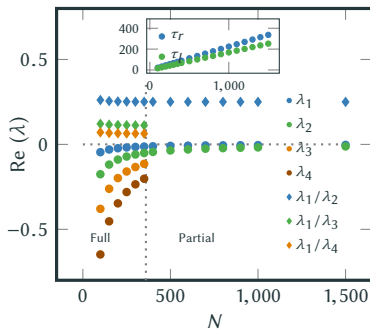
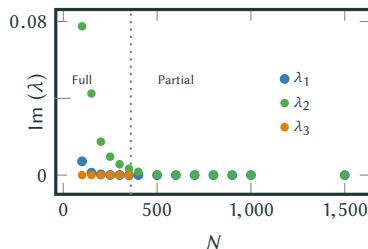
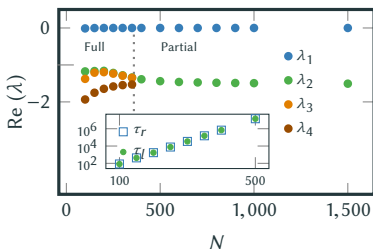
- Formal solution of the finite-size master equation reads

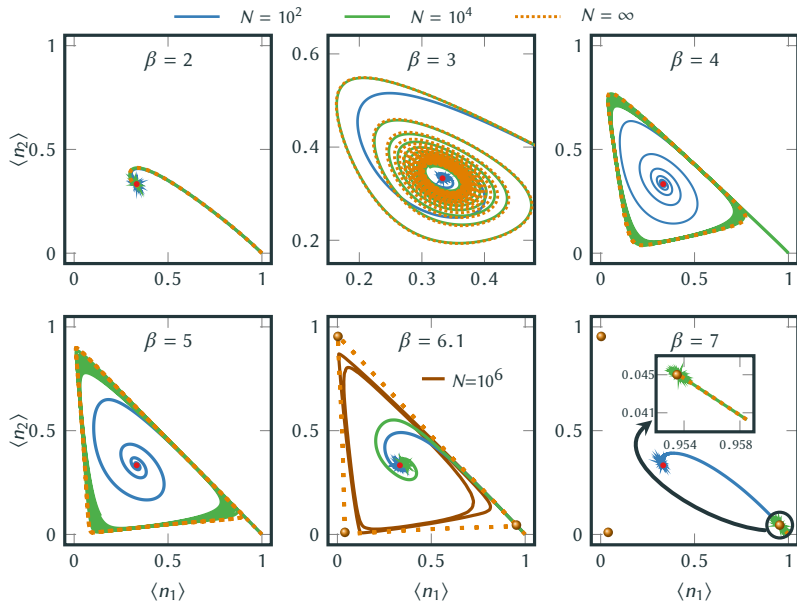
$$\mathbf{P}(t) = e^{\mathbf{W}t} \mathbf{P}(0) = \sum_{i, i^*} \underbrace{\Phi_i^L \mathbf{P}(0)}_{\equiv c_i} e^{\lambda_i t} \Phi_i^R + c_{i^*} e^{\lambda_{i^*} t} \Phi_{i^*}^R,$$

- Relevant **timescales** are specified by the slowest decaying modes.

Relaxation $\tau_r \sim -\frac{1}{\text{Re}(\lambda_1)}$, Metastable $\tau_m \sim -\frac{1}{\text{Re}(\lambda_2)}$, Lifetime $\tau_l \equiv \tau_r - \tau_m \sim \frac{\text{Re}(\lambda_1) - \text{Re}(\lambda_2)}{\text{Re}(\lambda_1) \text{Re}(\lambda_2)}$,



$\beta = 4.0$ (SM phase) $\beta = 7.0$ (AM phase)

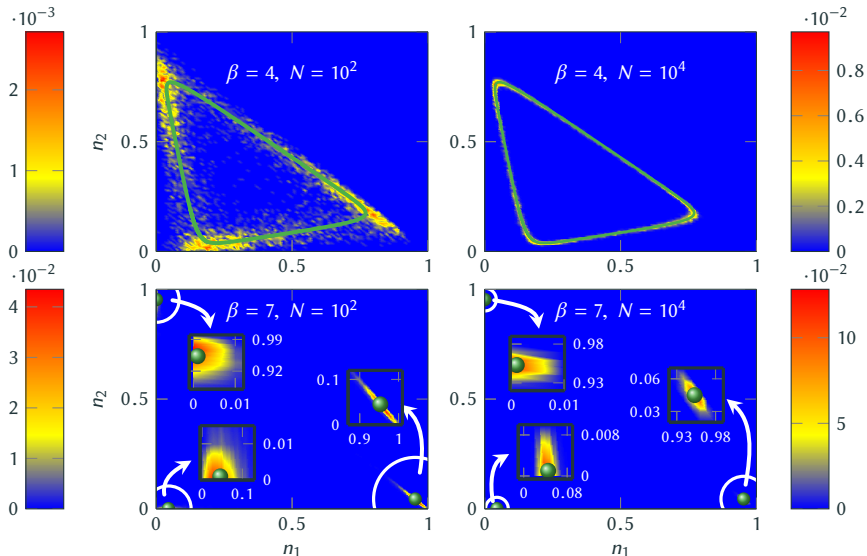


$\Rightarrow$ Large system dynamics indeed agrees with the mean-field solution at metastable times.

ROLE OF INITIAL CONDITION FOR METASTABILITY I

- Recall that $\mathbf{P}(t) \stackrel{t \gg \tau_m}{\approx} c_0 \Phi_0^R + e^{\lambda_1 t} c_1 \Phi_1^R + e^{\lambda_1^* t} c_1^* \Phi_1^{R*}$ with $c_{1,1^*} = \Phi_{1,1^*}^L \cdot P(0)$.
- Metastability can be removed from the dynamics for $P(0) = b_0 \Phi_0^{R*} + \sum_{i \neq 1, 1^*} b_i \Phi_i^{R*}$.

Is metastability a generic property of our system or merely an artifact of special initial conditions?



$$N = 10^2$$

$$\beta = 4$$

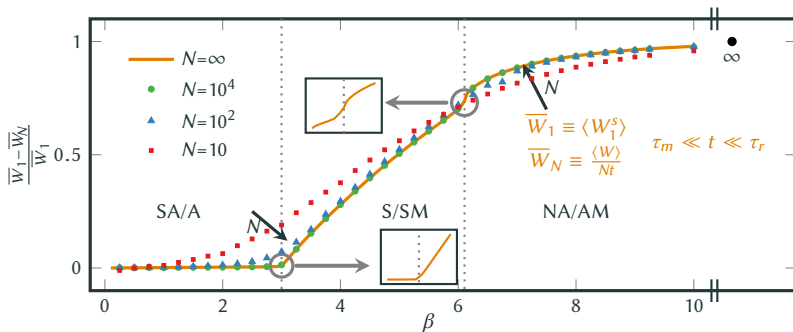
$$N = 10^4$$

$$N = 10^2$$

$$\beta = 7$$

$$N = 10^4$$

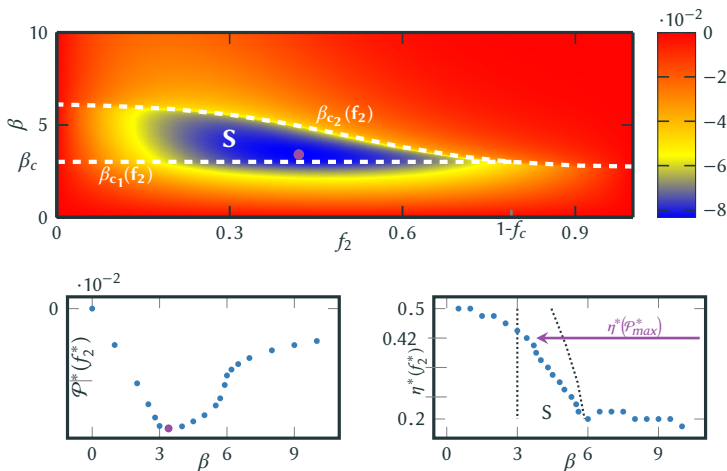
- **First Law:** $d_t \langle E \rangle = \langle \dot{Q} \rangle + \langle \dot{W} \rangle$
- With this setup, work needs to be done on average to maintain the system, $\langle \dot{W} \rangle > 0$, which is on average dissipated as heat into the bath, $\langle \dot{Q} \rangle < 0$.



- **(Signatures of) Nonequilibrium phase transitions** at the critical points (for large but finite N).
[“First-order phase transition” at β_{c1} and “second-order phase transition” at β_{c2}].
- **Reduction** of the dissipated work for **finite interactions** $u < 0$.
- Dissipated work is **further reduced** for **smaller networks** in the two high-temperature regimes.

EFFICIENCY AT MAXIMUM POWER

- **Work-to-work conversion** $f \equiv f_1 - f_2$, with input work $\mathcal{W}_1(f_1)$ and output work $\mathcal{W}_2(f_2)$.
- **Maximization of the output power** $\mathcal{P} \equiv \dot{\mathcal{W}}_2$ with respect to output force $\partial_{f_2} \mathcal{P}|_{f_1=1} = 0 \Rightarrow f_2^*$.
- Efficiency of work-to-work conversion at maximum power $\eta^* \equiv 1 - \left. \frac{f}{f_1} \right|_{f_2=f_2^*}$



- Power output is maximized in the **synchronization** phase.
- The **efficiency at maximum power**, $\eta^*(f_2^*)$, is surprisingly **close** to the universal value, $\eta^* = 1/2$, for a tightly-coupled (single net-current) system responding linearly.

- Minimal thermodynamic model exhibiting synchronization.
- Model with two non-equilibrium and one equilibrium phase transition.
- Resolving the apparent paradox of a finite irreducible Markov process and mean-field dynamics displaying stable oscillations and multistability via notions of metastability.
- Reduction of the dissipated work for finite interactions. For high and intermediate temperatures, smaller networks are favorable.
- When operating as work-to-work converter, the maximum output is achieved for a synchronous operating mode. The associated efficiency of this far-from-equilibrium scenario is close to linear-response value.

- Minimal thermodynamic model exhibiting synchronization.
- Model with two non-equilibrium and one equilibrium phase transition.
- Resolving the apparent paradox of a finite irreducible Markov process and mean-field dynamics displaying stable oscillations and multistability via notions of metastability.
- Reduction of the dissipated work for finite interactions. For high and intermediate temperatures, smaller networks are favorable.
- When operating as work-to-work converter, the maximum output is achieved for a synchronous operating mode. The associated efficiency of this far-from-equilibrium scenario is close to linear-response value.

⇒ [Herpich, Thingna, & Esposito arXiv:1709.06045]

- Minimal thermodynamic model exhibiting synchronization.
- Model with two non-equilibrium and one equilibrium phase transition.
- Resolving the apparent paradox of a finite irreducible Markov process and mean-field dynamics displaying stable oscillations and multistability via notions of metastability.
- Reduction of the dissipated work for finite interactions. For high and intermediate temperatures, smaller networks are favorable.
- When operating as work-to-work converter, the maximum output is achieved for a synchronous operating mode. The associated efficiency of this far-from-equilibrium scenario is close to linear-response value.

⇒ [Herpich, Thingna, & Esposito arXiv:1709.06045]

- Fluctuations? Local interactions? Topology? Genuine thermodynamic machine in synchrony?

- Minimal thermodynamic model exhibiting synchronization.
- Model with two non-equilibrium and one equilibrium phase transition.
- Resolving the apparent paradox of a finite irreducible Markov process and mean-field dynamics displaying stable oscillations and multistability via notions of metastability.
- Reduction of the dissipated work for finite interactions. For high and intermediate temperatures, smaller networks are favorable.
- When operating as work-to-work converter, the maximum output is achieved for a synchronous operating mode. The associated efficiency of this far-from-equilibrium scenario is close to linear-response value.

⇒ [Herpich, Thingna, & Esposito arXiv:1709.06045]

- Fluctuations? Local interactions? Topology? Genuine thermodynamic machine in synchrony?

Thank you for your attention!